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# Ambient Systems: Company Background

- Incorporated in 2021 in Singapore (2024 in Canada)

  -off from National University of Singapore (NUS)
- Received investment of $100k from NUS deep tech investment arm
- Received awards and recognition from Google Cloud and Panasonic for its product and innovation
- Seeking to develop a business planning software minimum viable product (MVP) in four industries


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# Project Objective: MVP Development

- Develop MVP for business planning software in transportation industry

  current MVP, cloud-based software

  - Create a demo for three related industries to demonstrate scalability
- Demonstrate plug-and-play real-time data collection and visualization capability using off-the-shelf sensor and cloud IoT service

  current machine learning models, hardware, and IP


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# Current Status

| | Use Case #1: Vertical Farming | Use Case #2: Decarbonization Platform for Commercial Real Estate | Use Case #3: Transportation |
|---|---|---|---|
| Status | Successfully developed a proof-of-concept for vertical farming in Singapore, incorporating multi-modal sensors and machine learning and secured an exclusive patent license for the invention. | Won first runner-up at a Google Cloud hackathon in Singapore for our innovative cloud-based AI solution for decarbonization in the commercial real estate sector. | Emerged as the winner of 2023 Tech Innovation Challenge, paving the way for a collaboration to leverage Panasonic's fast data transfer SW and 60 GHz V2X expertise in the transportation industry. |
| Current Components | • Hardware Sensors and Network<br>• ML model and patent<br>• Proprietary data | • Cloud-based platform<br>• ML model<br>• Open Source Academic Data | • System diagram |
| MVP: Business Planning Software | | | |

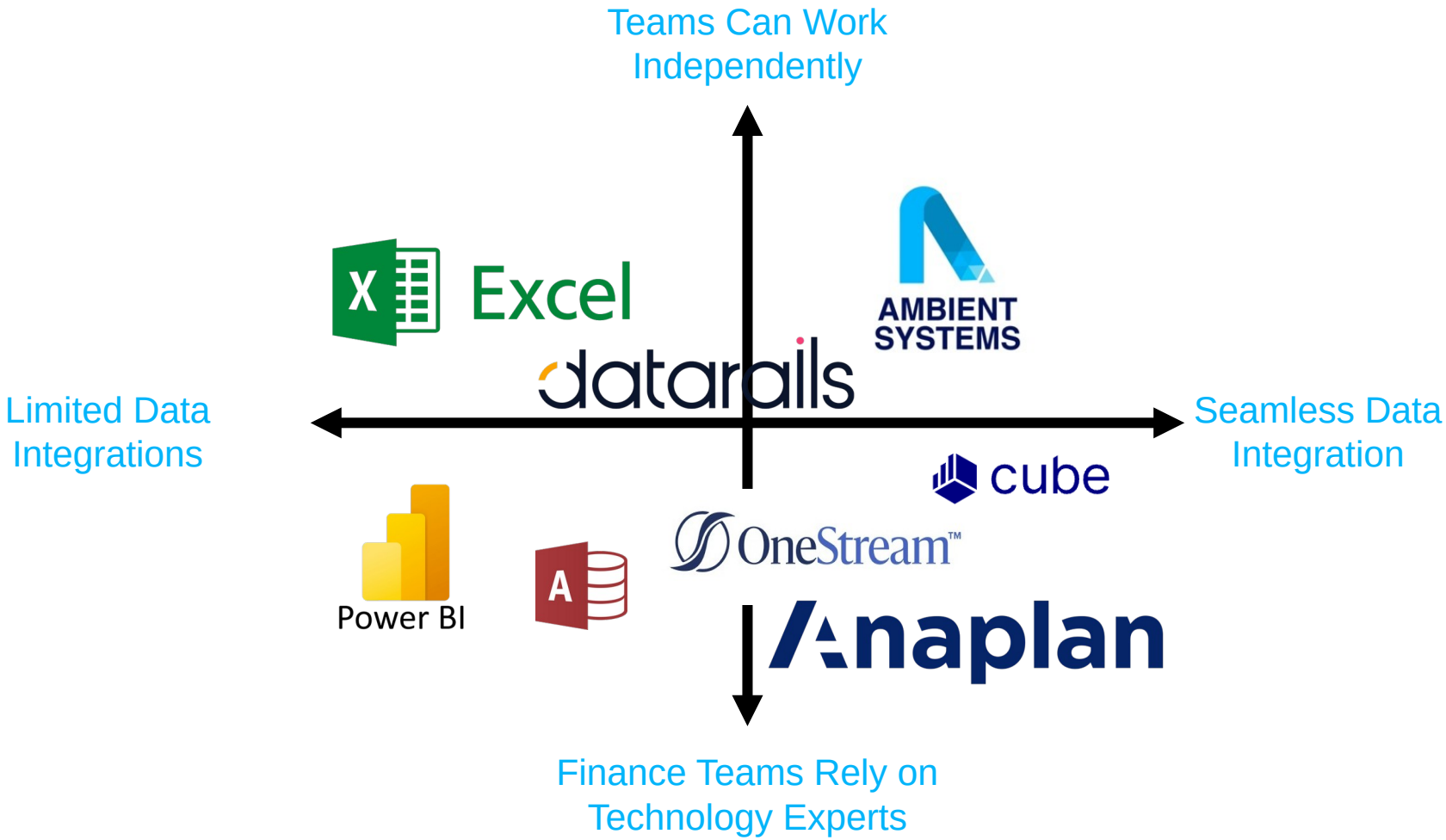

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# ERP Challenges

- Enterprise Resource Planning (ERP) software is often used to conduct business planning.
- However, ERPs are notoriously expensive and cumbersome to implement.
- Users often find them clunky and frustrating.
- Large companies spend millions of dollars annually to maintain and upgrade these systems.
- **Small companies cannot access them at affordable costs.**


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# Business Planning Software Market





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# Competitor on Features

| | Feature | Ambient Systems | Excel | DataRails | Access | PowerBI | Anaplan | Cube | OneStream |
|---|---|---|---|---|---|---|---|---|---|
| 1. Application | Cloud-Based | ✓ | X | X | X | X | ✓ | ✓ | ✓ |
| | Cross-Platform (Mac, Linux, Windows) | ✓ | | | X | X | Unsure | Unsure | Unsure |
| 2. Data Integration | Finance Data | | | | | | | | |
| | ERP Data | | | Unsure | Unsure | Unsure | | | Unsure |
| | Images & Videos | ✓ | X | X | X | X | X | X | X |
| | Documents | ✓ | X | Unsure | X | X | Unsure | Unsure | Unsure |
| 3. Finance | Calculates Metrics | ✓ | X | Unsure | X | X | Unsure | Unsure | Unsure |
| | Financial Analysis | ✓ | X | ✓ | X | X | ✓ | | |
| 4. Reporting | Dashboards Visualization | | | | | | | | |
| | PDF Reports | ✓ | X | Unsure | Unsure | Unsure | Unsure | Unsure | Unsure |
| 5. AI Capabilities | Custom AI Models | ✓ | X | Unsure | X | X | Unsure | Unsure | Unsure |
| | AI Assistant | ✓ | X | X | X | X | X | X | X |
| | Seamless Data Integration | ✓ | X | ✓ | X | X | Unsure | Unsure | Unsure |


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# Ambient Systems

Ambient Systems will provide a cloud-based, cross-platform application that integrates finance data, ERP data, images, videos, and documents. It will offer comprehensive finance features including metric calculation and financial analysis, along with robust visualization capabilities such as dashboards and PDF reports. Additionally, it will incorporate AI capabilities including custom AI models, an AI assistant for automatic report generation, and seamless data integration.


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# User Personas

| | | | |
|---|---|---|---|
| | | | **ona** |
| Role | Usually planning and forecasting. | Usual for costs and compliance. | Overs ng fleet management, routing, and logistics. |
| Responsibilities | Focuses on financial planning and analysis, including forecasting revenues and expenses, budgeting, and providing insights to support strategic decision-making. | Primarily concerned with managing costs, ensuring compliance with regulatory requirements and internal policies, and maintaining accurate financial records. | Ensures efficient vehicle and driver utilization while managing routing and scheduling to minimize transit times and costs, and coordinates with suppliers to optimize freight movement. |
| Needs | Requires tools and features that enable accurate forecasting, scenario planning, budget management, and the ability to analyze financial data to support business objectives. | Requires tools and features that streamline cost management processes, facilitate compliance with regulatory standards, automate financial reporting tasks, and provide visibility into financial data for auditing and reporting purposes. | Requires real-time tracking and monitoring tools for fleet management, alongside insights for route optimization, fuel savings, and operational efficiency. |

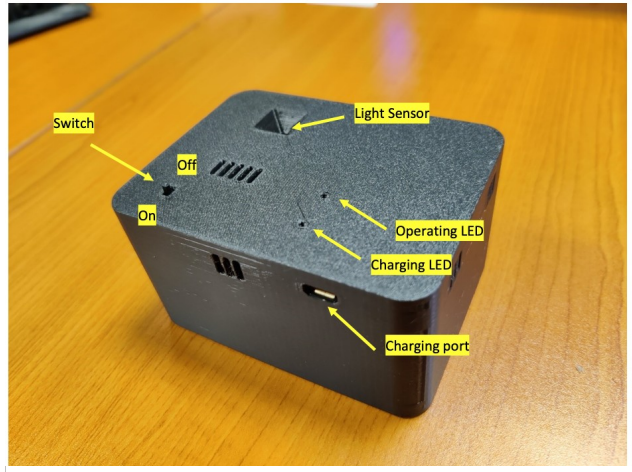

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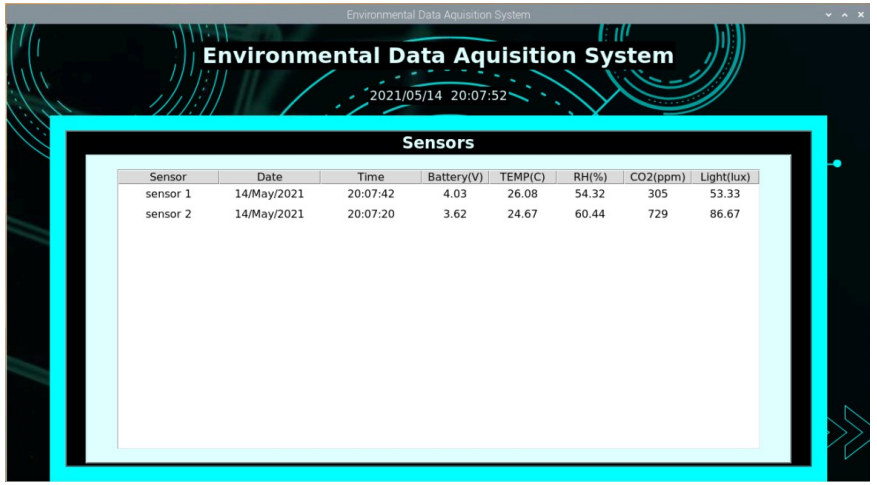
# User Stories

| Finance Teams | Ops Teams | Ops and Finance | Data |
|---|---|---|---|
| • Finance teams can forecast revenues and expenses.<br>• Finance teams can prepare reports for securing long-term funding and investments.<br>• Finance teams can calculate and minimize tax.<br>• Finance teams can reduce costs.<br>• Finance teams can manage cash flows and revenues transparently<br>• Finance teams can identify and manage risks.<br>• Finance teams can ensure compliance with regulations.<br>• Finance teams can analyze financial data to for decision-making. | • Operations teams can translate metrics/goals/KPIs into operational tasks.<br>• Operations teams can map operational tasks into financial goals.<br>• Operations teams can identify operational drivers.<br>• Operations teams can prioritize and sequence key value-added tasks.<br>• Operations teams can define task objectives and success criteria.<br>• Operations teams can use data to communicate and align. | • Ops and finance teams can align goals and objectives between finance and operations.<br>• Ops and finance teams can see the relevant data for each of our work tasks.<br>• Ops and finance teams can collaborate on analysis and decision making.<br>• Ops and finance teams can do a continuous improvement of the process.<br>• Ops and finance teams can partake in shared accountability and performance management.<br>• Ops and finance teams can utilize communication tools for seamless collaboration. | • Finance teams can access data from different systems.<br>• Finance teams can get data from operations using IoT. |

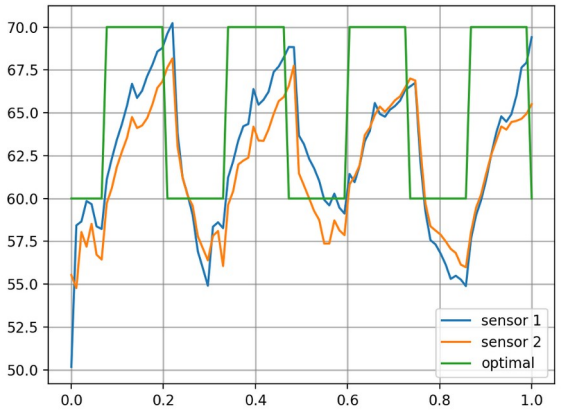

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# MVP #1: Vertical Farming





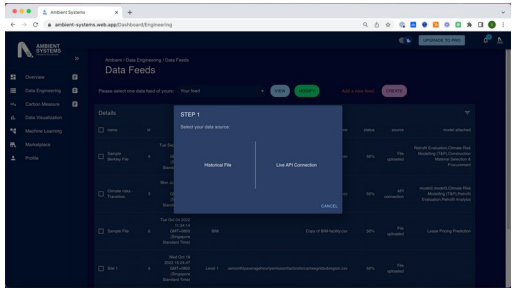





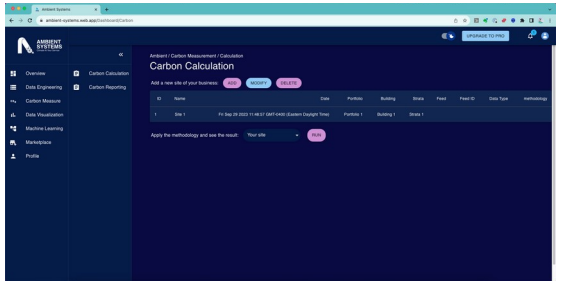


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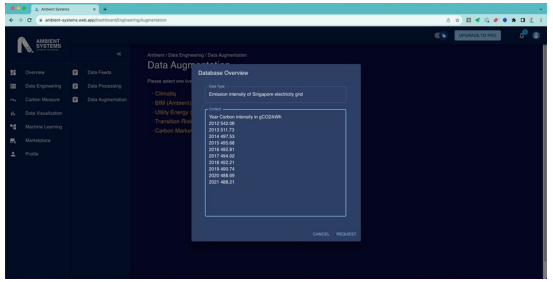
User Journey: Feature 1 – Combine Datasets



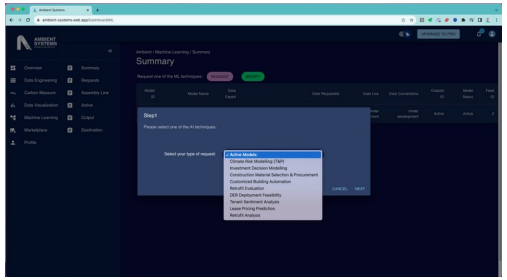

User Journey: Feature 2 – Automate Workflow



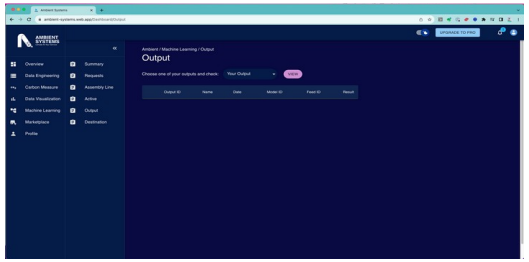

User Journey: Feature 3 – Add New Data




User Journey: Feature 4 – Leverage ML and AI




User Journey: Step 5 – Output




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# MVP #2: Commercial Real Estate

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# MVP #3: Transportation (to be developed)

- Two units:
  - One connected to vehicle
  - One connected to ‘collection point’ with access to wi-fi and or data
- Data from vehicle:
  - Telematics
  - In-vehicle camera recordings (door-facing)
    -facing camera recordings
  - Operational and financial data (e.g., fuel)
  - In-vehicle camera recordings (safety and security)
  - Bus driver-facing and front-facing camera recordings
- Device Assembly
- Vehicle-Device Attachments


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# MVP #4: Construction (to be developed)

- Data collection requirements to be established.


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# with Features

| 1. Cloud-Based & Cross-Platform Application | 2. Data Integration | 3. Finance Features | 4. Financial Analysis & Reporting | 5. AI Integration |
|---|---|---|---|---|
| • Develop a web application accessible from any device (Mac, Linux, Windows) and deployed on a (GCP/OVH/AWS). | • Develop data integration features to handle:<br>  • Finance data (e.g., income statements, balance sheets)<br>  • ERP data (e.g., inventory, customer data)<br>  • Images, videos, and documents<br>• Enhanced data integration feature with status, source, quality, and end-use highlights<br>• Seamless and automated data integration with validation checks<br>• Real-time plug-and-play data sensor integration with server (AWS/GCP/OVH) | • Financial metrics library focused on transportation<br>• Implement functionalities for calculating key financial metrics relevant to the transportation industry (e.g., fuel cost analysis, route optimization) | • Implement basic and advanced financial analysis tools<br>• Design and develop dashboards for visualizing financial and operational data<br>• Implement functionalities for generating PDF reports summarizing financial data and analysis | • Machine learning pipeline configuration for testing existing custom AI models in the platform<br>• Develop an AI Assistant to support users with tasks like report generation and data analysis<br>• AI Data Integration features to help user with input validation for correctness and completeness based on data mappings |


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